17th November, 2024

UNIT 4

L1 Key Management & Distribution

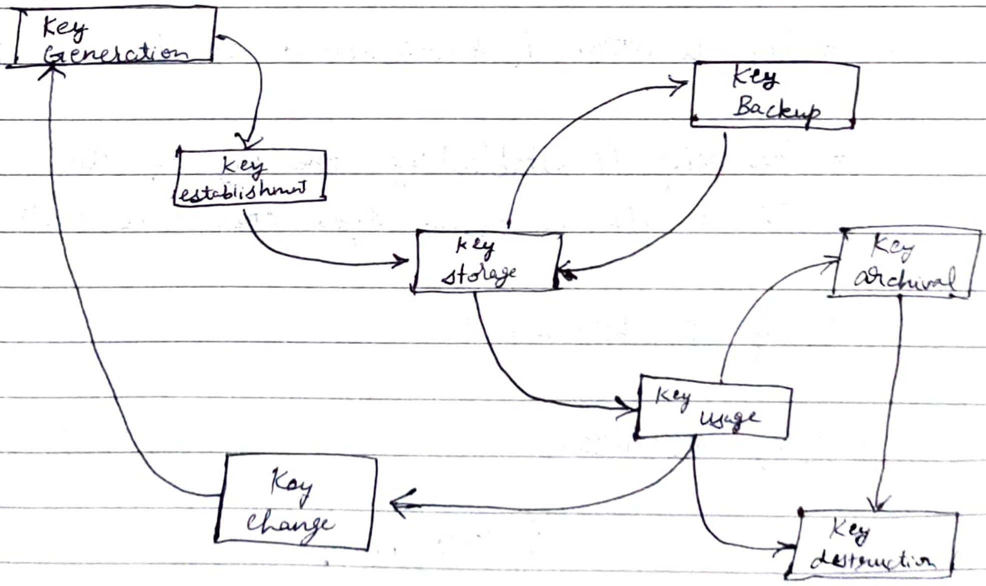
Kerckhoffs' Principle : the security of a cryptosystem should not rely on the secrecy of any of its workings, except for the value of the secret key

* Key agreement
* Key distribution

→ Key Management Fundamentals

1. Technical controls
2. Process controls – Policies, practices, procedures
3. Env controls
4. Human factors

→ Key Life Cycle



→ Considerations for Key Management

1. NW topology
2. Cryptographic mechanisms
3. Compliance restrictions
4. Standards
5. Many Organizations

* Key length changes with time

→ Key Generation

* Symmetric keys are just randomly generated nos.

HW based non-deterministic generator | SW based non-deterministic generator

→ Key Derivation

* Key generation & establishment is an expensive process
* Generating & establishing one key & then using it to derive many keys can save costs

→ Key Establishment

→ Key Hierarchy

$\boxed{\text{Master Keys}}$

↓

$\boxed{\text{Key encrypting Keys}}$

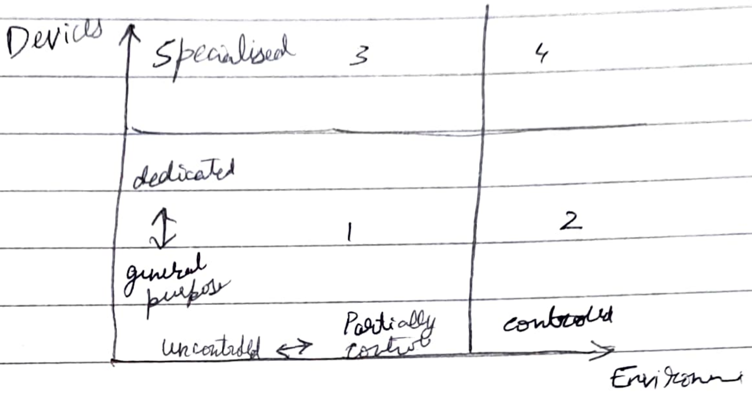
↓

$\boxed{\text{Data keys}}$

→ Key Storage

Devices
- General Purpose
- Dedicated
- Specialised

Environments
- Uncontrolled
- Partially controlled
- Controlled



17th November, 2024

Kerberos

21st November, 2024

Birthday Attack

$P(N)$ = Prob of atleast 2 people sharing a bday

$$P(N) = \left(\frac{1}{365}\right)^N \cdot \frac{365!}{(365-N)!}$$

for any $N \geq 366$ $P(N) = 100\%$
because of Pigeon Hole Principle

$$P(N \text{ people having diff bdays}) = \frac{365!}{((365-n)! * 365^n)}$$

→ Hash func. requirements

1. IP of variable length
2. OP fixed length
3. $H(x)$ is relatively easy to compute for any given x
4. $H(x)$ is one-way
5. $H(x)$ is collision-free.

→ Mechanics of Birthday Attack

1. The Precomputation Stage
2. The Collision Search Stage

→ Implications of Birthday Attack

1. Digital signatures
2. Password Storage System
3. SSL/TLS certificate

→ Best Practices

1. Adequate Hash Lengths
2. Robust Mathematical Foundation
3. Frequent Evaluation and Updates

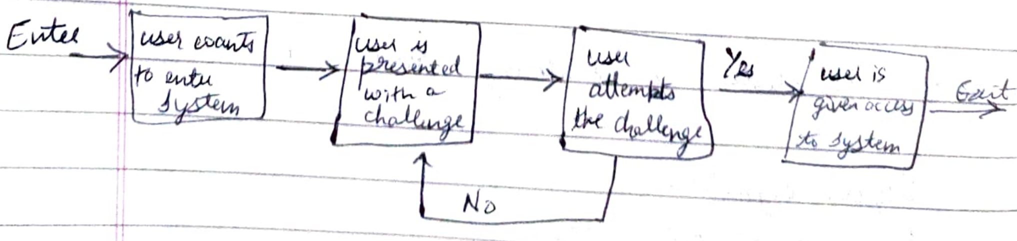
Entity Authenication methods

→ Common Authenication Types

1. Password based
2. Multi-factor
3. Certificate-based
4. Biometric
   - Facial
   - Fingerprint
   - Speaker
   - Iris
5. Token based

→ Challenge Response Authenication

Protocol has 2 components: question, response



→ CRAM
- Static
- Dynamic

→ Examples

1. CAPTCHA
2. Password
3. Biometrics
4. Salted Challenge Response Authenication Mech. (SCRAM)
5. SSH
6. Password Proof System
7. Challenge - Handshake Authenication Protocol
8. OATH Challenge-Response Algorithm
9. YubiKey
10. MD5

Zero Knowledge Proofs

→ Methods of entity auth:
1. Using a password
2. Challenge-response protocols
3. Zero-Knowledge protocols

→ Properties
1. Completeness
2. Soundness
3. Zero Knowledge

→ Types
1. Interactive
2. Non-Interactive

* Zero Knowledge Proof is a 3 pass identification protocol.

1. Commitment / witness from $A \to B$
2. Challenge $B \to A$
3. Final message $A \to B$

→ Types

1. Fiat - Shamir Protocol
2. Feige - Fiat - Shamir Protocol
3. Guillou - Quisquater Protocol

→ Fiat - Shamir Protocol

Alice
Secret S
random r

$$\text{Alice} \xrightarrow{x = r^2 \bmod N} \text{BoB}$$
$$\text{Alice} \xleftarrow{e \in \{0,1\}} \text{BoB}$$
$$\text{Alice} \xrightarrow{y = r * S^e \bmod N} \text{BoB}$$

① Setup : Before msg exchange

s : secret

n : composite, prod of 2 primes, public

Prover → Verifier

$V = (s * s) \bmod n$

② Commitment

$$\text{Prover} \xrightarrow{x} \text{Verifier}$$

$x = (r * r) \bmod n$

③ Challenge

$$\text{Prover} \xleftarrow{b} \text{Verifier}$$

④ Response

Prover $\xrightarrow{\quad y \quad}$ Verifier

$y = (r * s^b) \bmod n$

⑤ Verification

Prover

Verifier

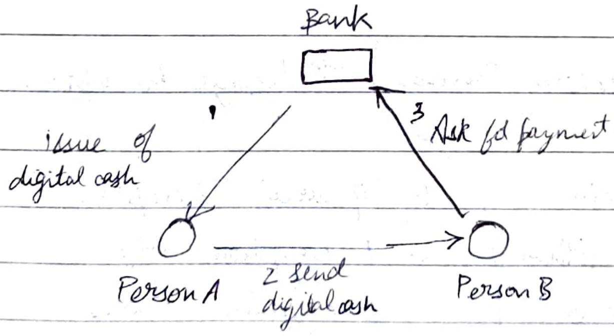
$y^2 \bmod n = x * v^b \bmod n$

→ Zero Knowledge is not completely anonymous as public keys are uses

- Pros : 1. Secured
  2. Simple

- Cons : 1. Limited
  2. Lengthy
  3. Imperfect

ECash



→ Cryptographic Description

1. One way func
2. Key Pairs
3. Signature & Identification
4. Secure Hashing

WhatsApp

→ Encryption (E2EE)

Public key from server

Private key decrypts message

User A – Public Key → WhatsApp Server → User B

encrypts message

→ Benefits of E2EE

1. Confidentiality
2. Protection Against Surveillance
3. Preventing data breaches
4. Trust in communication services
5. Personal Safety
6. Preserving user autonomy
7. Legal & Ethical compliance
8. Mitigating insider Threads
9. Global standard for privacy

→ Message Encryption Process

1. Data Preparation
2. Encryption Algorithm
3. Encryption Process
4. Transmission

→ Message Decryption Process

1. Ciphertext Reception
2. Decryption Key
3. Decryption Process
4. Message Presentation

→ Limitations of E2EE

1. Key Management
   - Key Distribution
   - Key Loss
2. No central Recovery

3. Limited MetaData Protection
4. User Errors
5. Incompatibility
6. Backups
7. Legal & Regulatory Challenges
8. Misuse and Abuse

→ Potential Advancements

1. Quantum-Resilient Encryption
2. Homomorphic Encryption
3. Post-Quantum Cryptography
4. Zero-Knowledge Proofs
5. Improved Key Management
6. Federated Learning & Encrypted AI
7. Decentralized & Blockchain-Based Encryption

HTTPS working

1. HTTPS Connection Request
2. Public Key Transmission
3. Session Key Generation
4. Session Key Encryption
5. Switch to Symmetric Encryption

Blockchain

- Cryptographic hash funcs
- Digital signatures
  - Enhanced security
  - Decentralization support
  - Traceability

→ Quantum Computing Threat

1. Shor's Algo : Asymmetric Algo's at Risk
2. Grover's Algo : Reducing security of Symmetric Cryptography

→ Families of Post-quantum crypto methods

1. Lattice-based algos
2. Code-based algos
3. Hash-based algos
4. Isogeny-based algos
5. Multivariate algos

→ Challenges of PQC

1. Lack of Standardization
2. Interoperability
3. Performance
4. Adoption
5. Cost

Date
Page

YT # Elgamal Cryptography

Asymmetric Key cryptography

① Key Generation
② Encryption
③ Decryption

→① Key Generation

1. Select large prime no. p $(p = 11)$
2. Select dec. key (private key) $(d = 3)$
3. Select 2nd part of enc key (e_1) $(e_1 = 2)$
4. Calculate 3rd part of enc key (e_2) $\boxed{e_2 = e_1^d \bmod p}$

$$e_2 = 2^3 \bmod 11$$
$$= 8$$

5. Public key $= (e_1, e_2, P)$ & pvt key $= d$
Public key $= (2, 8, 11)$

→② Encryption

1. Select random int (R) $(R = 4)$
2. Calculate c_1

$$c_1 = e_1^R \bmod p$$
$$= 2^4 \bmod 11 = 16 \bmod 11 = 5$$

$\boxed{c_1 = 5}$

3. Calculate c_2

assume $PT = 7$

$$c_2 = (P.T \times e_2^R) \bmod p$$
$$= 7 \times 8^4 \bmod 11$$
$$= 6$$

$\boxed{c_2 = 6}$

4. CipherText $(c_1, c_2) = (5, 6)$

→ ③ Decryption

$$PT = [c_2 \times ((c_1)^D)^{-1}] \mod p$$

$$= 6 \times (5^3)^{-1} \mod 11$$

$$= 6 \times 125^{-1} \mod 11$$

$125x \mod 11 = 1$

for $x = 3$ $\quad 375 \mod 11 = 1$

$\Rightarrow \boxed{x = 3}$

$\Rightarrow 6 \times 3 \mod 11 = 18 \mod 11$

$= 7$

$\Rightarrow \boxed{P.T = 7}$

Y7 # MD5 (Message Digest 5)

Fast & produces 128 bit message digest

→ Working of MD-5

① Padding

(extra bits)

Original message + padding

(⇒ tota len = 64 bits less than exact multiple of 512)

eg: Original msg = (1000 bits) + padding

$512 \times 1 = 512$

$512 \times 2 = 1024$

$512 \times 3 = 1536$

$$\begin{array}{r} 1536 \\ -\ 64 \\ \hline 1472 \end{array}$$

⇒ Total length = 1472

$$\begin{array}{r} 1472 \\ -1000 \\ \hline 472 \end{array}$$

⇒ padding = 472 bits

② Appending

Append the original length before padding

length mod 64 $\quad$ 1000 mod 64

(mostly 64 bits is answer)

∴ append 64 bits to 1472

⇒ Becomes multiple of 512

③ Divide

2nd step OP

512 bit block1 $\quad$ 512 bit block2 $\quad$ . . . . . $\quad$ 512 bit block n

④ Initializing (4 chaining variables)

each = 32 bit

Ⓐ Ⓑ Ⓒ Ⓓ values predefined

⑤ Processing 512 bit blocks

a) copy of variables

$A = a$, $B = b$, $C = c$, $D = d$

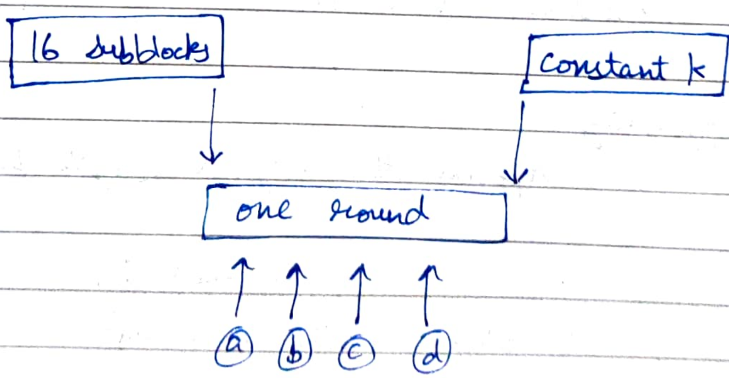
b) divide 512 bit blocks into 16 → 32 bit blocks

16 blocks → 32 bits each

$16 \times 32 = 512$

c) 4 rounds

16 subblocks & a constant k



$a = b + ((a + \text{Process}, P(b, c, d) + m[i] + T[k]))$

SHA Algorithm

modified ver of MD5

MD5 : OP len = 128 bit

SHA : OP len = 160 bit

→ Working

1. Padding (same of MD5)
2. Appending (same of MD5)
3. Divide IP into 512 bit blocks (same of MD5)

4. Initialize 5 chaining variables (A, B, C, D, E)

5. Process Blocks

A = a, B = b, C = c, D = d, E = e

divide into no. of 512 bit blocks

↓

16 sub blocks each 32 bit

4 rounds (each round = 20 steps)

RSA eg: PT = 9

$p = 7$
$q = 11$

$n = p \times q = 77$

$1 < e < \phi(N)$

$\phi(n) = (p-1)(q-1)$
$= 6(10)$
$= 60$

③ choose e coprime of 60 ~~77~~

N, $\phi(N)$ 77, 60

$e = 7$

$\langle e, n \rangle = \langle 7, 77 \rangle$

④ $C = m^e \bmod n$
$= 9^7 \bmod 77$
$= 4782969 \bmod 77$
$= 37$

⑤ pvt key

$de \bmod \phi(N) = 1$
$7d \bmod 60 = 1$

$7d \bmod 60 = 1$

$d = \frac{1 + K\phi(N)}{e}$

$d = \frac{1 + k(60)}{7} = 43$

$K = 0, 1, 2 \ldots$

(6) $M = c^d \bmod n$

$= 37^{43} \bmod 77$

$43 = (101011)_2$

1	0	1	0	1	1
37	60	67	23	15	9

$37^2 \bmod 77 = 60$

$60^2 \bmod 77 = 58$

$58(37) \bmod 77 = 67$

$67^2 \bmod 77 = 23$

$23^2 \bmod 77 = 67$

$37 \times 67 \bmod 77 = 15$

$15^2 \bmod 77 = 71$

$37 \times 71 \bmod 77 = 9$

Modular Arithmetic

eg: 23^3 mod 30

23^3 mod 30 = -7^3 mod 30 || 23 mod 30

= -7^3 mod 30

= $-7^2 \times -7$ mod 30

= 49×-7 mod 30

= -133 mod 30

= -13 mod 30

= 17 mod 30

= 17

q: 31^{500} mod 30

31 ÷ 30

remainder = 1

=> 1^{500} mod 30

1 mod 30

= 1

q: 242^{329} mod 243

-1^{329} mod 243

= $-1^{328} \times -1^1$ mod 243

= 242 mod 243

= 242

q: 11^7 mod 13

0·285714

q: $240^{262} \bmod 14$

$(262)_{(10)} = 100000110_{(2)}$

1st	2nd	3rd	4th	5th	6th	7th	8th	9th
1	0	0	0	0	0	1	1	0
240	4	2	4	2	4	4	4	2

$240^2 \bmod 14 = 4$

$4^2 \bmod 14 = 2$

$2^2 \bmod 14 = 4$

$4^2 \bmod 14 = 4$

$2^2 \bmod 14 = 4$

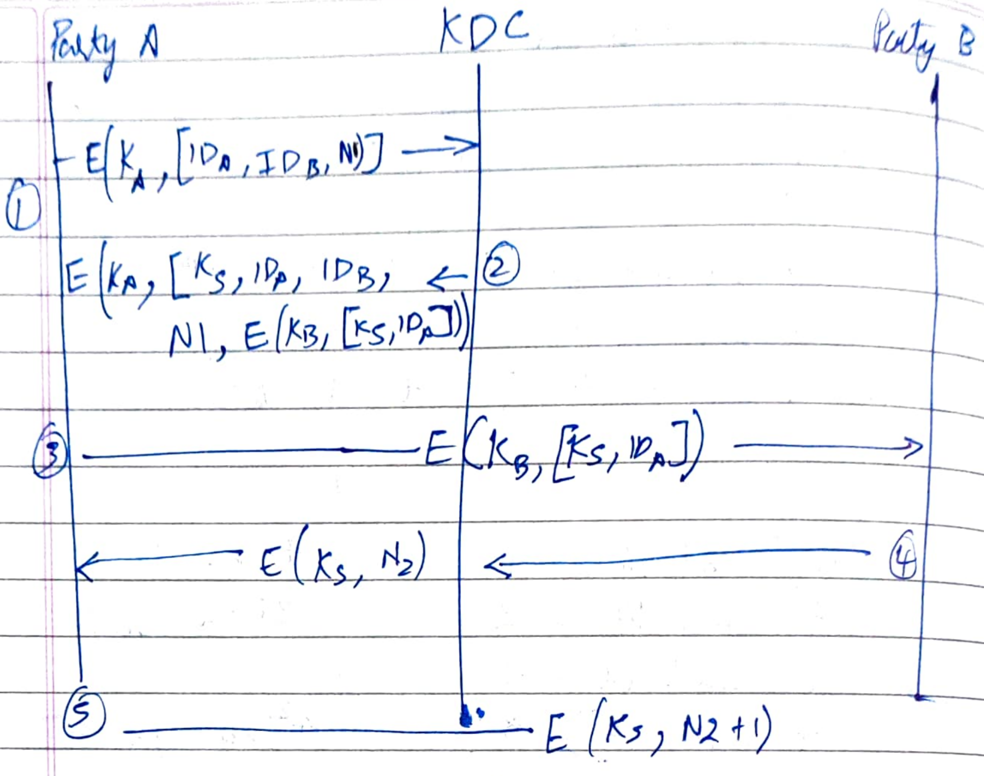
$4^2 \bmod 14 = 2$

$2 \times 240 \bmod 14 = 4$

$4^2 \bmod 14 = 2$

$2 \times 240 \bmod 14 = 4$

$4^2 \bmod 14 = 2$



TGT
- Session key
- User's ID
- Expiration time

15th December, 2024

Hill cipher

enc : $C = KP \mod 26$

dec : $P = K^{-1}C \mod 26$

a	b	c	d	e	f	g	h	i	j	k	l	m	n	o	p	q	r	s	t	u	v	w
0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19			

q. Key Matrix $= \begin{bmatrix} 2 & 3 \\ 3 & 6 \end{bmatrix}_{2\times2} = K$

PT = "attack"

Sol. $\begin{bmatrix} a \\ t \end{bmatrix}_{2\times1} = \begin{bmatrix} 0 \\ 19 \end{bmatrix}$ $\begin{bmatrix} t \\ a \end{bmatrix} = \begin{bmatrix} 19 \\ 0 \end{bmatrix}$ $\begin{bmatrix} c \\ k \end{bmatrix} = \begin{bmatrix} 2 \\ 10 \end{bmatrix}$

i) $\begin{bmatrix} a \\ t \end{bmatrix} = \begin{bmatrix} 0 \\ 19 \end{bmatrix}$

$C = KP \mod 26$

$C = \begin{bmatrix} 2 & 3 \\ 3 & 6 \end{bmatrix}_{2\times2} \begin{bmatrix} 0 \\ 19 \end{bmatrix}_{2\times1} \mod 26$

$C = \begin{bmatrix} 57 \\ 114 \end{bmatrix} \mod 26 = \begin{bmatrix} 5 \\ 10 \end{bmatrix} = \begin{bmatrix} f \\ k \end{bmatrix}$

ii) $\begin{bmatrix} t \\ a \end{bmatrix}$

$C = \begin{bmatrix} 2 & 3 \\ 3 & 6 \end{bmatrix} \begin{bmatrix} 19 \\ 0 \end{bmatrix} \mod 26$

$= \begin{bmatrix} 38 \\ 57 \end{bmatrix} \mod 26 = \begin{bmatrix} 12 \\ 5 \end{bmatrix} = \begin{bmatrix} m \\ f \end{bmatrix}$

iii) $\begin{bmatrix} c \\ k \end{bmatrix}$

$C = \begin{bmatrix} 2 & 3 \\ 3 & 6 \end{bmatrix} \begin{bmatrix} 2 \\ 10 \end{bmatrix} \mod 26$

$= \begin{bmatrix} 34 \\ 66 \end{bmatrix} \mod 26 = \begin{bmatrix} 8 \\ 14 \end{bmatrix} = \begin{bmatrix} i \\ o \end{bmatrix}$

CT = fkmfio

$12 - 9 = 3 \quad \frac{1}{3}\begin{bmatrix} 6 & -3 \\ -3 & 2 \end{bmatrix}$

$C_1 = \begin{bmatrix} f \\ k \end{bmatrix} = \begin{bmatrix} 5 \\ 10 \end{bmatrix} \quad C_2 = \begin{bmatrix} m \\ f \end{bmatrix} = \begin{bmatrix} 12 \\ 5 \end{bmatrix} \quad C_3 = \begin{bmatrix} i \\ o \end{bmatrix} = \begin{bmatrix} 8 \\ 14 \end{bmatrix}$

$PT = K^{-1} C \mod 26$

$K = \begin{bmatrix} 2 & 3 \\ 3 & 6 \end{bmatrix}$

$\cancel{K^{-1} = \begin{bmatrix} 2 & -1 \\ -1 & 2/3 \end{bmatrix}}$

$\cancel{\begin{bmatrix} 2 & -1 \\ -1 & 2/3 \end{bmatrix}\begin{bmatrix} 5 \\ 10 \end{bmatrix} = \begin{bmatrix} 0 \\ \end{bmatrix}}$

$10 - 10$

$-5 + \frac{20}{3}$

$K = \begin{bmatrix} 2 & 3 \\ 3 & 6 \end{bmatrix} \quad adj(K) = \begin{bmatrix} 6 & -3 \\ -3 & 2 \end{bmatrix}$

$|K| = 3$

$K^{-1} = \frac{1}{|K|} adj(K)$

$DD^{-1} = 1 \mod 26$

$3(D^{-1}) = 1 \mod 26$

$D^{-1} = \frac{1 + K(26)}{3} = 9$

$\therefore K^{-1} = D^{-1} \, adj K$

$= 9 \begin{bmatrix} 6 & -3 \\ -3 & 2 \end{bmatrix} =$

$9 \begin{bmatrix} 6 & 23 \\ 23 & 2 \end{bmatrix} = \begin{bmatrix} 54 & 207 \\ 207 & 18 \end{bmatrix} \mod 26$

$$K^{-1} = \begin{bmatrix} 2 & 25 \\ 25 & 18 \end{bmatrix}$$

$$PT_1 = K^{-1} C \mod 26$$

$$= \begin{bmatrix} 2 & 25 \\ 25 & 18 \end{bmatrix} \begin{bmatrix} 5 \\ 10 \end{bmatrix} \mod 26$$

$$= \begin{bmatrix} 260 \\ 305 \end{bmatrix} \mod 26$$

$$= \begin{bmatrix} 0 \\ 19 \end{bmatrix}$$